

Example 3

Referring again to Table 1.1, the offer side 5-year swap rate is 1.647% semi bond. A good approximation for the offer side 5-year swap rate quoted on a quarterly bond basis is found using Equation 1.8. We see that if the 5-year swap rate is 1.647% semi bond, then the 5-year swap rate quoted quarterly bond should be approximately

$$4 \times \left[\left(1 + \frac{.01647}{2} \right)^{2/4} - 1 \right] = 1.644\%.$$

And if the 5-year swap rate is 1.644% quarterly bond, then a decent approximation to the swap rate quoted on a quarterly money basis can be found using Equation 1.9. In particular, the rate should be approximately

$$1.644\% \times \frac{360}{365} = 1.621\%. \quad (1.10)$$

□

Bid and Offer Swap Rates

Once again, when we talk about swaps we almost always focus on the fixed side exclusively. And as mentioned previously, the bid side swap rate is defined as the fixed rate at which a swap dealer is willing to pay fixed, and the offer side swap rate is defined as the fixed rate at which a swap dealer is willing to receive fixed. When a client is receiving fixed, he wants to receive as high of a fixed rate as possible, and of course a dealer is incentivized to pay as low of a rate as possible. Similarly, when a client is paying fixed, he wants to pay as low of a fixed rate as possible, and the dealer is incentivized to receive as high of a rate as possible. As a convention, most composite pages show the bid and offer semi bond swap rates as the bid and offer swap spread over the mid-market Treasury. Although this serves as a very reasonable indication of where the swaps market is, it is often more appropriate to take the relevant bid or offer side Treasury yield into account. If the size of the trade is small enough to get done “on screen” (meaning at the levels that appear on the composite page), then

$$\begin{aligned} &\text{Bid Side Swap Rate} \\ &= \text{Offer Side Treasury} + \text{Bid Side Swap Spread} \end{aligned} \quad (1.11)$$

$$\begin{aligned} &\text{Offer Side Swap Rate} \\ &= \text{Bid Side Treasury} + \text{Offer Side Swap Spread}. \end{aligned} \quad (1.12)$$

This may seem a bit mysterious, so let's try to shed some light on it. Suppose that a dealer is asked to pay fixed in a swap by a client. Let's forget about the floating side right now and just focus on the fixed side.²⁹ If the dealer is paying fixed, thereby agreeing to pay out fixed coupons over the life of the swap, it is kind of like the dealer is selling a fixed rate bond to the client. Think about it: the client is receiving those fixed coupons. Therefore, the client's position, looking at the fixed leg of the swap in isolation, is very similar to having purchased a fixed rate bond from the dealer.³⁰

Let's remind ourselves now of a few facts about fixed rate bonds. If we own a fixed rate bond, and the bond market *rallies* (meaning rates go down), then the value of our bond goes up. Conversely, if we own a fixed rate bond and the bond market *sells off* (meaning rates go up), then the value of our bond goes down. Of course, if we are instead short the bond, then we benefit when the market sells off and are hurt when the market rallies.

Let's get back to the swap dealer who just paid fixed in a swap. As we suggested, by paying fixed it is as if the dealer has just sold a fixed rate bond. In fact, after entering into the swap and paying fixed, the dealer will be underwater on its position if the market rallies and will be making money if the market sells off. The dealer may want this exposure, or it may look to hedge itself to immunize against changes in interest rates.

One way the swap dealer could hedge himself is to buy Treasuries in the market.³¹ Let's keep in mind that swap dealers are market makers in swaps. They are not market makers in Treasuries. Thus, when the swap dealer goes to the Treasury market, he is a customer of that market and has to buy Treasuries on the offer side and sell Treasuries on the bid side. Since the swap dealer would have to buy Treasuries on the offer side if he went in to hedge against his paying fixed, it is more customary to regard the bid side swap spread as being spread over the offer side yield corresponding to the offer side price at which he could actually buy those Treasuries.

Figure 1.4 shows how this works. Let's suppose a swap dealer agrees to pay fixed in a 5-year swap to a client and goes to the Treasury market and

²⁹It turns out, as we will soon see, that we often ignore the floating side when we look at valuing swaps, so get used to it now. The floating side of the swap for purposes of valuation is like Rodney Dangerfield: it don't get no respect.

³⁰Obviously, one big difference is that a bond has principal that the holder is entitled to, but let's ignore that for now.

³¹It is not the case that the dealer is buying Treasuries in order to use coupon payments to have cash necessary to make the fixed coupon payments in the swap. The Treasuries are bought by the dealer in order to give him a position that has similar, but opposite, interest rate sensitivity to the swap. The Treasuries that are bought by the swap dealer will go up in value if the market rallies and down in value if the market sells off, which is the opposite of the change in the value of the swap due to a movement in rates, as we will see.

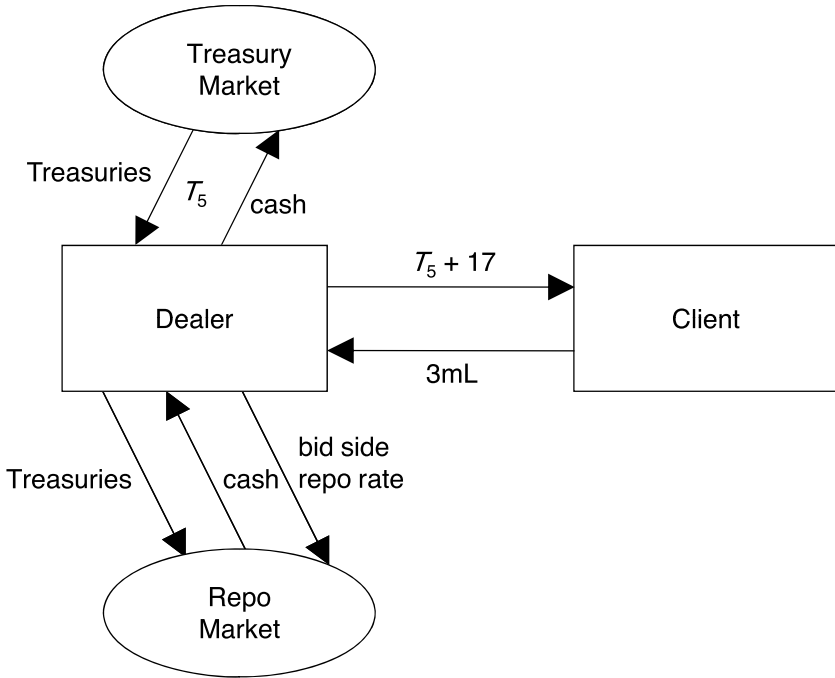


Figure 1.4: Mechanics Behind a Swap Dealer Paying Fixed and Buying Treasuries to Hedge

buys 5-year Treasury notes from a Treasury dealer at the offer side price of 101-16. The yield corresponding to this price is 1.436%. The fixed rate in the swap is, in accordance with Equation 1.11, the offer side yield of 1.436% plus the bid side swap spread of 17 bps, giving a bid side swap rate of 1.606%.

What's the big deal? This number is only a tenth of a bp different from what was shown on the composite page for the bid side swap rate. Well, suppose that the notional on the swap was really big like \$500mm, in which case the swap dealer would have to buy about \$500mm Treasuries to hedge himself.³² A \$500mm Treasury trade is a big trade. Maybe the price on the screen of 101-16 is good for \$25mm Treasuries. Suppose that if the dealer goes in to the Treasury market to buy another \$50mm, this will push the

³²The swap dealer would not necessarily have to buy exactly \$500mm Treasuries to hedge himself since the interest rate sensitivity of a swap (which we haven't talked about yet) per million notional is not generally equal to the interest rate sensitivity of a Treasury of comparable maturity per million principal.

price up to 101-16+. Maybe he can get another \$100mm there, but the next \$50mm he has to pay 101-17. In order to set the rate in the swap for a large trade like this, a swap dealer is going to estimate a weighted average dollar price that he is going to have to pay for the Treasury hedge. He will need to take into consideration that as he tries to accumulate enough Treasuries to hedge he will likely push the price up. Suppose he estimates that, given market conditions, he is going to have to pay a weighted average price of 101-19. The yield corresponding to this price is 1.417%. If the dealer agrees to pay fixed at 17 bps over where he can buy Treasuries, then in this case the bid side swap rate would be $1.417\% + 0.17\% = 1.587\%$. Notice here that the effect is to lower the swap rate at which the dealer is willing to pay. If the size of the trade was small and could get done “at screen levels,” then the rate the dealer would be willing to pay is 1.606%. But for this larger trade, when we incorporate where the swap dealer can actually buy his Treasury hedge, the rate the dealer would be willing to pay is nearly 2 bps lower than the level on the screen. As we will be able to show later in Chapter 2, the present value of this rate differential is on the order of magnitude of \$400,000.

Repo

This story (the one of a swap dealer who pays fixed and buys Treasuries to hedge) isn't quite over yet. Suppose a swap dealer pays fixed on \$100mm 5-year swaps and is able to buy \$100mm 5-year Treasuries at the price on the screen of 101-16 to hedge his position. The dealer is going to have to pay over \$100mm to the Treasury dealer for these Treasuries. Where is the swap dealer going to get this money? He probably doesn't have \$100mm lying around. What he will do is enter into a *repurchase agreement* (also known as a *repo*) in the repo market. A repo is similar to a secured loan. It is the sale of a security with the simultaneous agreement to repurchase the security at a predetermined price at some time in the future. Implicit in a repo is a *repo rate*, which is used to determine the price at which the securities will be repurchased. The counterparty on the other side of the trade who takes in the security in exchange for cash is known to have done a *reverse repo*. Repo markets exist for a variety of security types, including Treasuries and mortgages.³³

The length of a repo can be *overnight*, for *term* (meaning a specified period of time of more than one day), or *open* (meaning a nonspecified period of time of more than one day).³⁴ Term repos typically do not exceed one year.

³³For more information about the repo markets, see, for example, Stigum (2007), Fabozzi (2001), Sundaresan (2009), or Tuckman (2002).

³⁴In an open repo, there is no predetermined price at which the security will be repurchased, since the length of the contract has not been specified.

A popular time frame for a term repo of an on-the-run Treasury security is *to-the-date*, or to the settlement date of the next Treasury of the same maturity series that will be issued. If the securities should pay any coupons over the life of a repo agreement, the counterparty who delivered the securities in exchange for cash in the repo is entitled to the coupon interest.

Whereas an investor will often enter into a repo in order to fund a security purchase, his counterpart who enters into a reverse repo is doing so in order to earn interest on cash he has on hand. Oftentimes those who take in securities for cash in a reverse repo are indifferent as to the particular Treasury securities they take in. When the particular Treasury securities that are delivered in a repo are not specified, the securities are referred to as *general collateral*. Certain investors, such as swap desks who need to fund particular Treasuries that they have purchased, need to specify particular Treasuries in a repo. Such repos are referred to as *special*. The repo rate for specials is no greater than the repo rate on general collateral and is often significantly lower. When the availability of a particular security is relatively low, and the demand to take delivery of the particular security in a reverse repo is relatively high, this imbalance manifests itself by putting downward pressure on repo rates. Such Treasuries are said to be *trading special*. On-the-run Treasury securities typically trade special in the repo market.

Let's continue with our example of the swap dealer who pays fixed on \$100mm 5-year swaps and hedges the trade by buying \$100mm on-the-run 5-year notes at 101-16. This is the offer side price shown on the composite page in Table 1.1. The market data on this page is from August 11, 2010. The on-the-run 5-year Treasury on this date was the 1.75% of July 31, 2015. Treasuries purchased on this date would settle *reg*, or $T + 1$, on August 12, 2010. The *total invoice price* or *dirty price* paid by the buyer on the settlement date includes accrued interest. In this case, the dirty price for the bond is³⁵

$$\$100\text{mm} \times (101.5\% + 0.05706522\%) = \$101,557,065.22. \quad (1.13)$$

If the swap dealer goes to the repo market to enter into a repo on August 12, 2010 to fund this position for n days, then the swap dealer will deliver

³⁵For Treasuries, accrued interest is calculated on an Act/Act basis. In the case of the 1.75% of July 31, 2015 for settlement August 12, 2010, there are 12 days of accrued interest (the number of days between July 31, 2010, and August 12, 2010) and 184 days in the accrual period (i.e., the number of days between July 31, 2010, and January 31, 2011). Thus, the amount of accrued interest as a percentage as par is given by

$$0.875\% \times 12/184 = 0.05706522\%.$$

\$100mm face of the security and will receive \$101,557,065.22.³⁶ When the repo matures, the swap dealer will take delivery of the securities and owe a total of

$$\$101,557,065.22 \times \left(1 + \frac{n}{360} \times \text{repo rate}\right), \quad (1.14)$$

where the repo rate is the n -day rate of interest charged by a repo dealer for the use of its funds. For instance, if the repo is overnight (in which case $n = 1$) and if the repo rate is 20 bps, then the swap dealer will owe

$$\$101,557,065.22 \times \left(1 + \frac{1}{360} \times 0.20\%\right) = \$101,577,629.43.$$

When the swap dealer goes into the repo market to do a repo, it is a customer of the repo market and will have to transact at the *bid side repo rate* when taking in cash and delivering securities and at the *offer side repo rate* when providing cash and taking in securities.³⁷ Figure 1.4 summarizes the position of the swap dealer who pays fixed in a swap, buys Treasuries to hedge, and enters into an overnight repo to finance the Treasury purchase for a day.

Resets and Payments

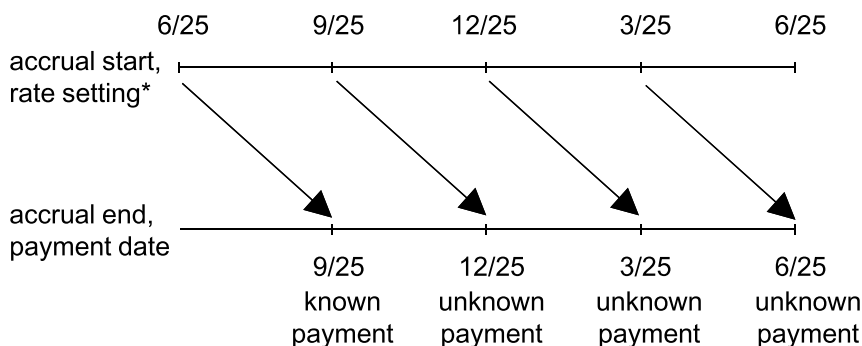
If we transact a USD swap, a *spot starting* transaction has an *effective date* (the date from which payments begin to accrue), which is “ $T + 2$,” or two New York and London Business Days after the trade date.³⁸ Suppose we transact a swap on a Tuesday. Then if Wednesday and Thursday are both Business Days (sometimes referred to as “*good days*”) in New York and London, then the effective date of the swap will be that Thursday. Suppose now we transact a swap on a Thursday. If Friday and Monday are good days, then the effective date of the swap will be Monday since Saturdays and Sundays are never good days. If, however, Monday is not a good day in either New York or London, then the effective date would be Tuesday.

The resets and payments made on the LIBOR leg of the swap typically occur with “resets in advance, payments in arrears.” This is demonstrated in

³⁶Often repos are subject to *haircuts* by dealers, which stipulate that the borrower receives an amount of cash lower than the value of the securities delivered as collateral. This protects the dealer (to some extent) in case the borrower defaults and the value of the pledged securities should fall, in which case liquidating the bonds would not make the dealer whole. We shall ignore haircuts for our purposes here.

³⁷Repo desks are market makers in repo and profit from the bid/offer spread between the bid side repo rate and the offer side repo rate. The bid side rate (at which repo desks earn interest) is higher than the offer side rate (at which they pay interest).

³⁸A *Business Day* for a particular city is defined as a day upon which commercial banks and foreign exchange markets are generally open to settle payments in the specified city.



* Rate setting is two London Business Days prior to start of accrual period.

Figure 1.5: Resets and Payments on the Floating Side

Figure 1.5. Suppose that we transact a spot starting swap on June 23, and it has an effective date of June 25. The first 3-month LIBOR setting will be on June 25, and that rate will accrue for the quarter, and payment will be made on September 25, assuming that is a good day. On September 25, LIBOR will be set for the next quarter, and that rate will accrue for the quarter, and that payment would normally be made on December 25. Since December 25 is a holiday in New York and London (i.e., neither commercial banks nor foreign exchange markets are open to settle payments on this day), neither a payment nor a LIBOR reset will occur on that day. This payment date falls on a “bad day” (i.e., a weekend or holiday). So when is this payment made? Payments in swaps can be made on a *Preceding* (meaning payments are made on the first good Business Day prior to the bad day), a *Following* (payments are made on the first good day after the bad day), or a *Modified Following* (payments are made on the first good day after the bad day, provided that does not bring us into the next month; if it does bring us to the next month, then that particular payment is made the first good day prior to the bad day) basis, which determines when a payment that would otherwise fall on a bad day is paid. For a USD swap, the standard convention is Modified Following.

Exactly how much should be paid when we need to move the payment date from a bad day depends on whether payments are *adjusted* (in which case the number of days that are accrued is altered in accordance with the payment date, and hence the magnitude of the payment for that period changes) or *unadjusted* (in which case the number of days that a rate is accrued for does not change despite changing the payment date). For a USD swap, the standard is that payments are adjusted. Appendix C goes into a lot

more detail about how these conventions come into play and gives an example demonstrating why one should be careful about which conventions apply.

One other slight technicality to be aware of: in a standard swap the LIBOR rate that is set on each reset date is actually the LIBOR setting determined by the BBA two London Business Days prior to each reset date. Thus, if we transact a swap on June 23 and it has an effective date of June 25, the LIBOR setting for the initial period uses the rate two London Business Days prior to June 25, which is June 23. Thus, for a spot starting swap, the initial LIBOR set for the first three-month period is typically known with certainty when entering into the trade. All other LIBOR rates will not be known until we get to the respective rate setting dates.

Sample Term Sheet

It used to be commonplace that a dealer would send a client a sample term sheet (or vice versa) prior to executing a swap in order to make sure that each counterparty was in agreement with the terms of the trade. These days, a term sheet is rarely sent prior to executing a relatively simple transaction but may be sent in advance of more complicated or exotic transactions. It is, however, very common today to send a quick term sheet (perhaps more aptly called a trade recap) soon after a swap has been transacted in order to review the terms to which the two counterparties have agreed. Whether or not a term sheet is used, a more formal confirmation is sent by one counterparty to the other after the trade has been executed. It is signed by both counterparties and is a legally binding agreement of the swap.³⁹ Figure 1.6 shows a sample term sheet. The last line of the confirm

³⁹What happens when the terms of the written confirmation do not match the terms of the trade that has been executed? This is not that uncommon, actually. Most often the two people who execute the trade are not the two people who draft the confirm, and sometimes, like a game of telephone, the written confirmation does not accurately reflect the trade that was executed. Assuming both counterparties agree that the signed confirmation does not accurately reflect the terms of the trade as executed, a revised written confirmation is signed by both counterparties.

But what if one of the counterparties says that the signed confirmation does not accurately reflect the terms of the trade as executed, but the other counterparty says the signed confirmation is accurate? That gets complicated. In this case, the two counterparties may try to reach a compromise. In the event that the two counterparties are unable to reach a compromise, a lawsuit may be necessary to settle the matter. The area of law that is relevant here is referred to as the *statute of frauds*. In the event that a voice trade and a written confirmation do not match, the voice trade is the trade (i.e., the oral contract is the binding contract). A counterparty who believes that a signed confirmation does not accurately reflect the terms of a trade at the point of execution will likely need to prove that is the case. In such cases, electronic records of the trades (e.g., telephone recordings) may provide such evidence, provided that, at a minimum, such recordings are legal. See, for example, Gooch and Klein (2002).

INDICATIVE TERM SHEET

Fixed Rate Payer:	Swap Dealer
Floating Rate Payer:	Client
Notional Amount:	USD 100,000,000
Trade Date:	August 11, 2010
Effective Date:	August 13, 2010
Maturity Date:	August 13, 2015
Fixed Rate Payment Dates:	The 13th day of each February and August, commencing February 13, 2011, up to and including the Maturity Date
Fixed Rate:	1.607% (= T_5 + 17)
Fixed Rate Day Count:	30/360
Floating Rate:	3-month USD-LIBOR-BBA
Floating Rate Determination Date:	2 London Business Days prior to each accrual start period
Floating Rate Payment Dates:	The 13th day of each November, February, May, and August, commencing November 13, 2010, up to and including the Maturity Date
Floating Rate Day Count:	Act/360
Documentation:	ISDA

Figure 1.6: A Sample Term Sheet

indicates that an ISDA will govern the trade. ISDA is an organization that, among other things, has created a legal document, known as an ISDA Master Agreement, that two counterparties who enter into swaps have typically signed in advance of transacting any OTC derivatives together. The Master Agreement carefully spells out certain legal conditions that govern swaps and gives definitions to many of the terms the counterparties may use in a confirmation.

OIS Swaps

An *Overnight Index Swap* (OIS) is a fixed-floating swap in which the floating leg is tied to a reference overnight rate. In the USD swaps market, that overnight reference rate is the Federal Funds effective rate. The fixed rate in an OIS is referred to as the *OIS rate*. In an OIS, one counterparty will pay the OIS rate, and the other will pay an average of the Federal Funds effective rate over the term of the swap. The OIS rate for a given maturity is often thought of as the expected Federal Funds rate over the term of the swap.

It has become common in the market to look at the *LIBOR-OIS spread*, the difference between a LIBOR rate and an OIS rate of the same maturity. Many in the market regard this spread as a proxy for the health of the financial system. The Fed Funds effective rate is the rate that banks lend to one another overnight in the Fed Funds market. The LIBOR-OIS spread is thought to measure the risk premium that banks charge to lend to one another above and beyond the funds rate. This interpretation comes from the fact that a bank could either roll funds over at the overnight rate for some period of time, say, three months, or, alternatively, lock in a rate at 3-month LIBOR. In times of stress, there may be a significant premium attached to lending funds for three months, compared to lending overnight in the Fed Funds market.

The LIBOR-OIS spread had been stable for many years but saw a tremendous widening at the height of the crisis. For example, as shown in Figure 1.7, the spread between 3-month LIBOR and 3-month OIS traded consistently around 10 bps prior to the crisis but widened out dramatically when Lehman fell, trading in excess of 300 bps at the height of the crisis. When LIBOR and OIS tracked each other very closely, there was not a tremendous amount of interest in OIS swaps. But since the time of the crisis, trading in OIS has picked up significantly. Investors have used the product to hedge against or take a view on unanticipated policy changes from the Fed.